Probabilistic Machine Learning

Block 1 – Introduction & Probabilistic Regression

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Outline

1. Introduction to Probabilistic Machine Learning
2. Probabilistic Linear Regression

Machine Learning Is Everywhere... But Can We Trust It?

ML models make predictions that affect **real decisions**:

A diagnostic model says
"*benign tumour*"

A credit model says
"*approve the loan*"

A self-driving car says
"*no obstacle ahead*"

A weather model says
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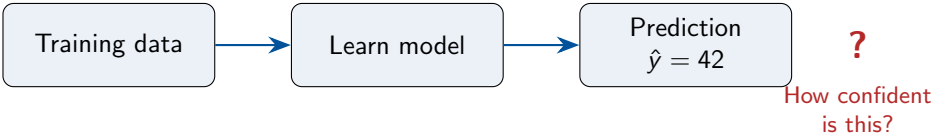
In every case, a single number is not enough.

What's Missing in Standard ML?

The **classical pipeline**:



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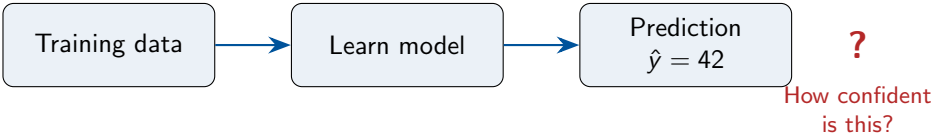

Standard ML models typically **cannot**:

- Express **confidence** — are all predictions equally reliable?
- Say **"I don't know"** — what happens in regions with no data?
- Quantify **noise** — how much variability is inherent in the problem?

Standard ML gives us **answers**, but not how much to **trust** them.

The Probabilistic Perspective — Intuition

Classical approach:

"What will the temperature be tomorrow?"

$\longrightarrow$ **22°C**

A single number.
Take it or leave it.

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Probabilistic approach:

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probably between 20–24°C.

A **distribution**
over possible answers.

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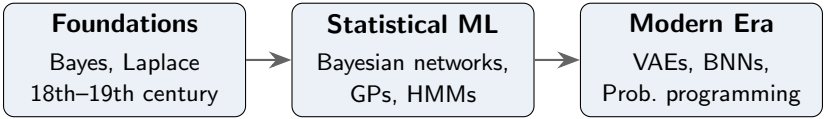
Core idea: Instead of predicting a single value,
predict a **probability distribution** over all possible values.

Probabilistic ML in Context

Probabilistic thinking in ML is **not new** — but modern tools have made it practical:



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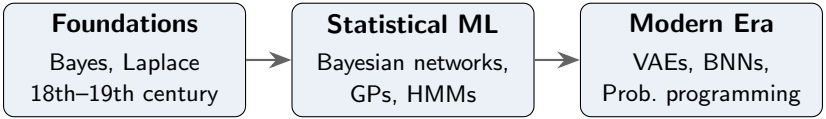

Today: languages like **Pyro**, Stan, and NumPyro let us:

- Write models in a few lines of Python.
- Perform inference **automatically** using gradient-based methods.
- Scale to large datasets and deep neural networks.

This course: learn the **ideas** and the **tools** to put them into practice.

What Does a ML Model Return?

When we train a model, we typically get a **point prediction**:

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"The price of this house is €250,000."

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But how much should we **trust** this prediction?

- Is it equally reliable for all inputs?
- What if the model has never seen similar data?
- Would we make the same decision regardless of confidence?

From Functions to Distributions

Classical ML: learn a function

$$x \longrightarrow \hat{y} = f(x)$$

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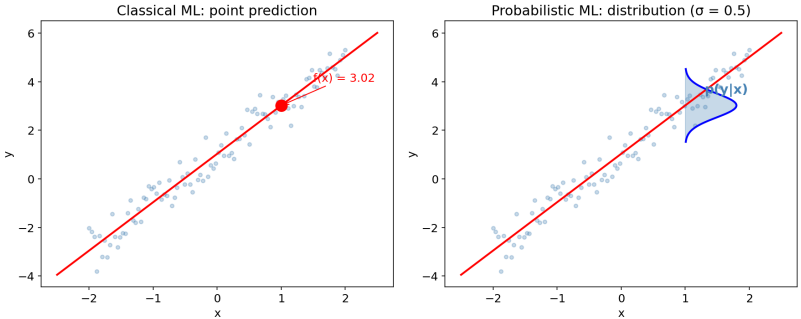
Probabilistic ML: learn a *conditional distribution*

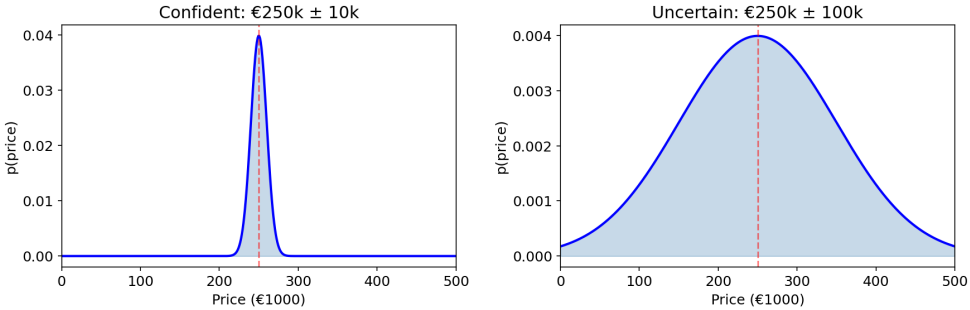
$$x \longrightarrow p(y \mid x)$$

Not a single answer, but a **distribution over possible answers**.



Why Uncertainty Matters

Two predictions with the **same mean** but very different uncertainty:



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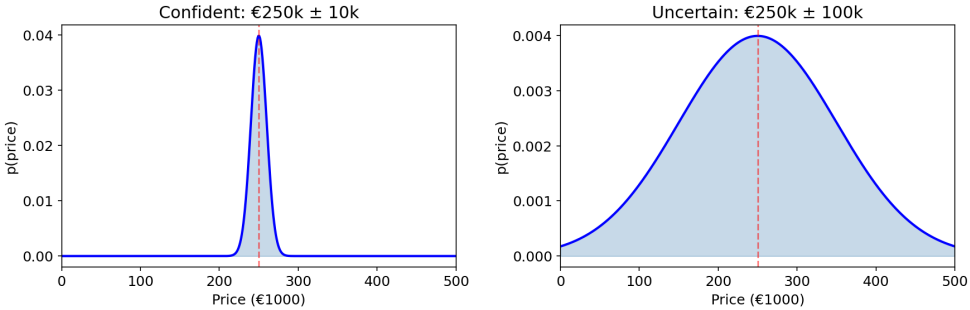

- **Left:** "€250k ± 10k" — confident, act on it.
- **Right:** "€250k ± 100k" — uncertain, be cautious.

Same prediction, very different decisions.

When Does Uncertainty Matter Most?

Noisy data
Observations are inherently imprecise.

Sparse regions
Few training points $\Rightarrow$ high uncertainty.

Ambiguous inputs
Multiple valid outputs for the same input.

Risk-sensitive decisions
Medicine, autonomous driving, finance...
The *cost of being wrong* varies.

Probabilistic ML lets us distinguish confident from uncertain predictions.

A Unifying Framework

Many classical algorithms are **special cases** of probabilistic models:

Classical method	**Probabilistic model**	**Distribution**
Linear regression	Gaussian linear model	$\mathcal{N}(\mathbf{w}^\top \mathbf{x} + b,\ \sigma^2)$
Logistic regression	Bernoulli model	$\mathrm{Bernoulli}(\sigma(\mathbf{w}^\top \mathbf{x} + b))$
k-means	Gaussian mixture model	$\sum_k \pi_k \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$
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These are not isolated algorithms — they are **instances of the same framework**.

Decoupling Modeling and Inference

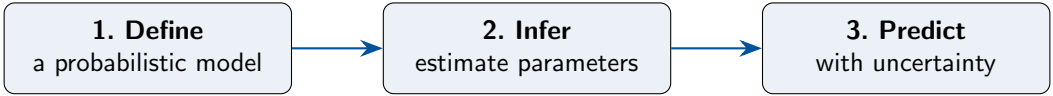
In classical deep learning these are often **tightly coupled**.

In probabilistic ML they are **separate concerns**:

1. Define a probabilistic model of the data.
2. Choose an inference method (MLE, variational inference, . . .).

PML precisely supports this separation.

The Probabilistic ML Recipe

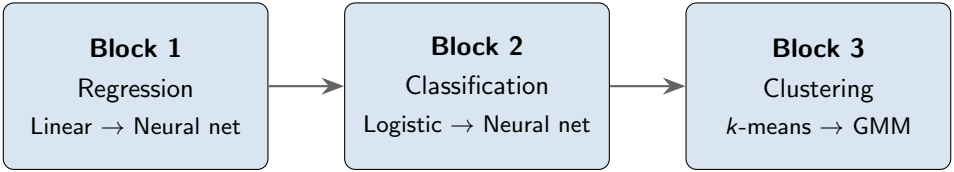


We will follow this recipe **three times today**:

1. **Block 1:** Regression
2. **Block 2:** Classification
3. **Block 3:** Clustering (unsupervised)

Same framework, same tool (Pyro), increasing complexity.

Today's Roadmap

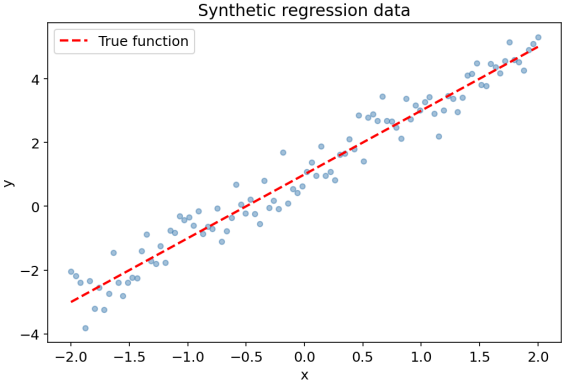


Each block:

1. Conceptual presentation
2. Pyro notebook (simple model)
3. Extension to more expressive models
4. Pyro notebook (extended model)

Regression: The Setup

Given pairs $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, we want to predict y from $\mathbf{x}$.



How should we model the relationship between $\mathbf{x}$ and y?

The Standard Approach

Fit a linear function minimizing **Mean Squared Error**:

$$\hat{\mathbf{w}}, \hat{b} \;=\; \arg\min_{\mathbf{w},b} \frac{1}{N} \sum_{i=1}^{N} \big(y_i - \mathbf{w}^\top \mathbf{x}_i - b\big)^2$$

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This works, but raises questions:

- Where does MSE come from? Why this loss and not another?
- How confident are we in the predictions?
- Can we quantify the noise in the data?

A probabilistic perspective answers all three.

A Probabilistic Model for Regression

Assumption: observations are noisy versions of a linear function.

$$y = f_\theta(\mathbf{x}) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

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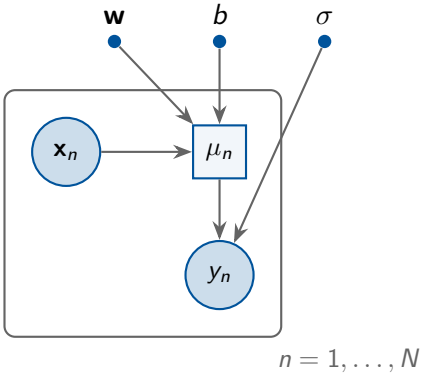
The model explicitly represents:

- the **signal**: $\mathbf{w}^\top \mathbf{x} + b$ (what we predict)
- the **noise**: σ^2 (how much variability remains)

Graphical Model (Plate Notation)

The probabilistic model can be represented as a **Bayesian network** using plate notation:



● = parameter / fixed value (`pyro.param`)

○ = observed r.v. (`pyro.sample`, `obs=y`)

▢ = plate (`pyro.plate`)

□ = deterministic function

$$\mu_n = \mathbf{w}^\top \mathbf{x}_n + b \qquad y_n \sim \mathcal{N}(\mu_n, \sigma^2)$$

Likelihood

Given the model, the probability of observing the data is:

$$p(\mathcal{D} \mid \theta) \;=\; \prod_{i=1}^{N} p(y_i \mid \mathbf{x}_i, \theta) \;=\; \prod_{i=1}^{N} \mathcal{N}\big(y_i \;\big|\; \mathbf{w}^\top \mathbf{x}_i + b,\; \sigma^2\big)$$

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This is the **likelihood function**: it measures how well the model with parameters $\theta = (\mathbf{w}, b, \sigma^2)$ explains the observed data.

Goal: find θ that *maximizes* the likelihood.

Maximum Likelihood $\Rightarrow$ MSE

Maximizing likelihood = minimizing **negative log-likelihood**:

$$
\begin{aligned}
-\log p(\mathcal{D} \mid \theta) &= -\sum_{i=1}^{N} \log \mathcal{N}\big(y_i \mid \mathbf{w}^\top \mathbf{x}_i + b,\ \sigma^2\big) \\
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The Punchline

Key insight

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It is the **consequence** of assuming Gaussian noise.

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Different distributional assumptions $\Rightarrow$ different loss functions:

Noise distribution	**Resulting loss**
Gaussian $\mathcal{N}(0, \sigma^2)$	Mean Squared Error (MSE)
Laplace $\mathrm{Laplace}(0, b)$	Mean Absolute Error (MAE)

The loss function is a **consequence** of the model, not an ad-hoc design decision.

σ^2 Is Also a Parameter

σ^2 is not just a technical detail — it is a **model parameter**:

- It is estimated jointly with $\mathbf{w}$ and b via MLE.
- It quantifies the **noise level** in the data.
- It determines the **width of our uncertainty bands**.

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Recall the NLL includes a term $\frac{N}{2} \log \sigma^2$:

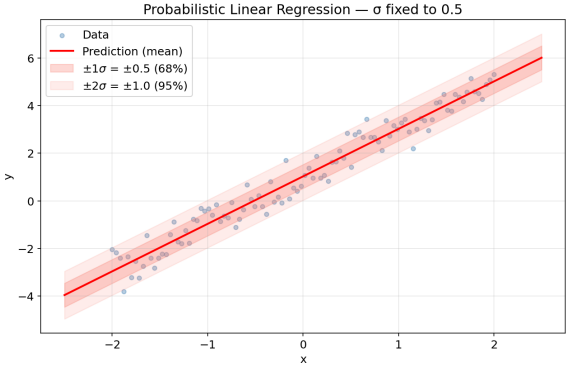
- Too small $\sigma \Rightarrow$ large penalty on residuals.
- Too large $\sigma \Rightarrow$ penalized by the log term.
- MLE finds the **right balance**.

(Extension: σ could depend on $\mathbf{x}$ — heteroscedastic noise. We won't implement this today, but Pyro supports it naturally.)

What We Gain

With a probabilistic model we obtain:

1. **Predictions** — the mean $\mathbf{w}^\top \mathbf{x} + b$.
2. **Uncertainty** — calibrated intervals via σ^2.
3. **Model comparison** — principled, via likelihood.



Preview: From Linear to Nonlinear

What if the data are **not linear**?

Replace the linear function with a **neural network**:

$$y \mid \mathbf{x} \sim \mathcal{N}\big(f_\theta(\mathbf{x}),\ \sigma^2\big) \qquad \text{where } f_\theta = \text{neural network}$$

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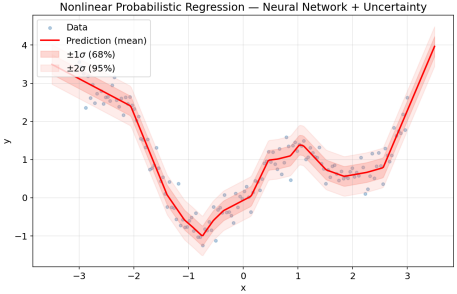
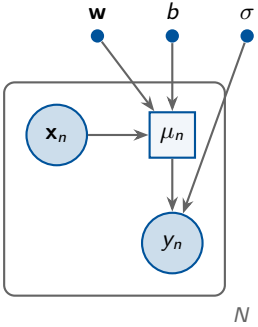
The probabilistic model does not change.
Only the **function family** of the predictor changes.



Same Graphical Model, Different Function

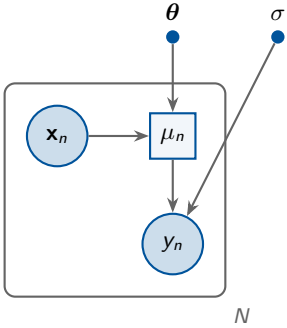
Linear regression



$$\mu_n = \mathbf{w}^\top \mathbf{x}_n + b$$

$\Rightarrow$

Neural network regression



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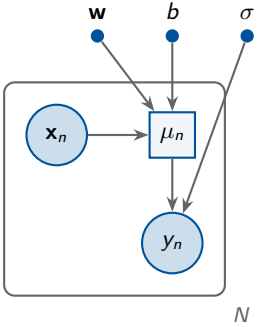

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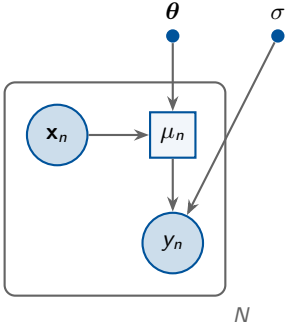

$$\mu_n = f_{\boldsymbol{\theta}}(\mathbf{x}_n)$$

The graphical model structure is identical. Only the deterministic function $\mu_n = g(\mathbf{x}_n)$ changes — the probabilistic layer ($y_n \sim \mathcal{N}(\mu_n, \sigma^2)$ inside the plate) remains the same.

Let's See It in Code

Time to open the **Pyro notebook**!

What we will implement:

1. Define the probabilistic linear regression model in Pyro.
2. Train via maximum likelihood (SVI).
3. Visualize predictions **with uncertainty bands**.
4. Compare with the nonlinear (neural network) extension.

The model you just saw on the slides translates *directly* into Pyro code.

Our Tool: Pyro

Pyro is a probabilistic programming language built on top of **PyTorch**.

- Developed by **Uber AI Labs** (now maintained by the open-source community).
- Lets you write probabilistic models in pure Python.
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Key idea: you already know PyTorch — Pyro adds a thin probabilistic layer on top.

https://pyro.ai